

1 Introduction

Many models can achieve equivalent performance on a given task while reaching that performance in different ways. This phenomenon is known as the **Rashomon Effect**.

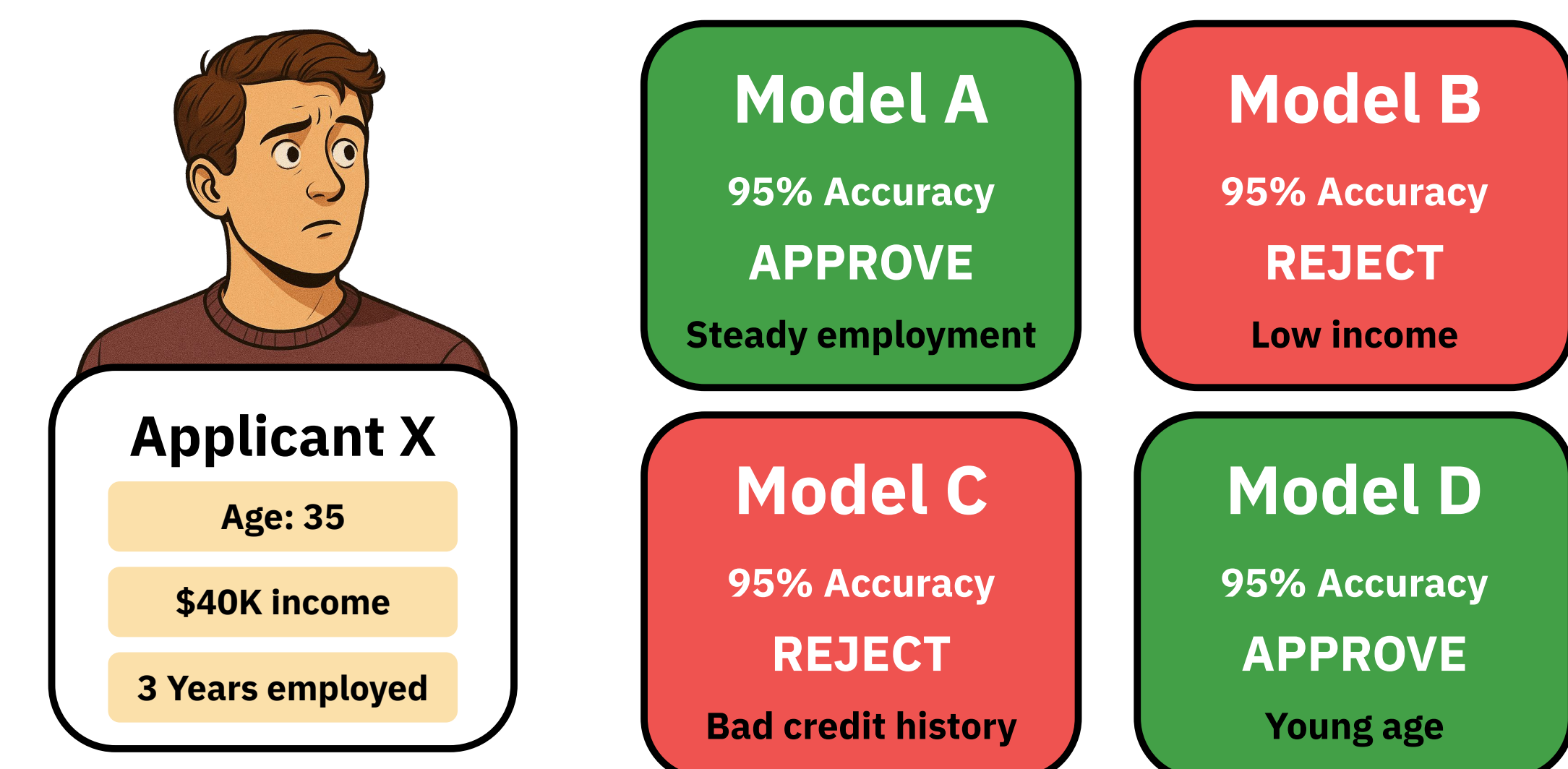
The Rashomon Set

The collection of these functionally diverse yet accurate models is the **Rashomon set**. Within it, models may produce divergent predictions for identical inputs. As the complete Rashomon set is computationally infeasible, we use an **empirical Rashomon set**.

$$\mathcal{R}_\epsilon^m = \{f_{w_i} \in \mathcal{H}_m \mid \hat{\mathcal{R}}(f_{w_i}) \leq \hat{\mathcal{R}}(f_{ref}) + \epsilon\}$$

Why Does This Matter?

Rashomon sets capture hidden model flexibility, supporting **fairness**, **uncertainty estimation** and **interpretability**. Yet equally accurate models may assign different outcomes to the same person, being a direct threat to trust in high-stakes AI systems for end users.



Four equally accurate models, one applicant, four different decisions. Which one is right?

Key Contributions

- Gradient-free Rashomon Set Exploration**
FiLM + CMA-ES, no retraining or gradient access required
- Validated Across Datasets and Architectures**
MLP, CNN and ViT on MNIST, CIFAR-10 and PneumoniaMNIST
- Competitive Diversity, Lower Cost**
Outperforms dropout on diversity; retraining on speed

2 Methodology

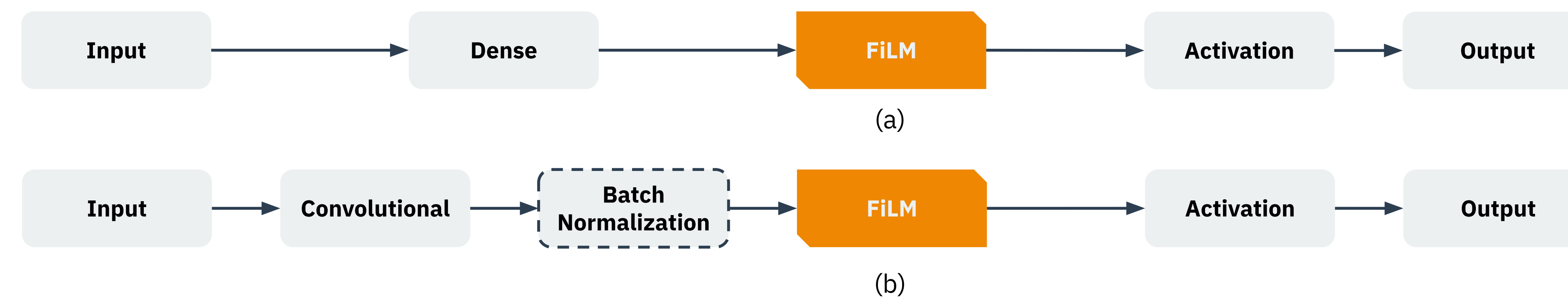
1. Train the Reference Model

Standard supervised learning until validation performance stabilises. All weights are then **frozen**. Reference loss defines the Rashomon threshold.

2. Wrap With FiLM Modulation Layers

Feature-wise Linear Modulation (FiLM) layers are inserted at certain positions in the reference model. A single shared latent vector $\mathbf{z} \in \mathbb{R}^d$ controls all layers simultaneously. At $\mathbf{z} = \mathbf{0}$, the original model is recovered.

$$FiLM(h; \mathbf{z}) = \gamma(\mathbf{z}) \odot h + \beta(\mathbf{z})$$



FiLM layer placement for an MLP (a) and a CNN (b), inserted after Batch Normalization when present (dotted). All FiLM layers share a single latent $\mathbf{z}$ -vector (orange).

3. Search Modulation Space With CMA-ES

CMA-ES is a **gradient-free evolutionary strategy** that iteratively adapts a Gaussian distribution over the search space, sampling candidate $\mathbf{z}$ -vectors and updating the distribution toward regions of higher fitness. To guide the search, the fitness function balances two objectives: **maximising disagreement** while **staying within the Rashomon tolerance**.

Diversity Score

$$\text{Div}_\lambda(\mathbf{z}) = \lambda \cdot \text{TVD}(p_{\text{ref}}(x), p_{\mathbf{z}}(x)) + (1 - \lambda) \text{Dis}(f_{\text{ref}}, f_{\mathbf{z}})$$

$\lambda = 0.5$ balances soft (TVD) and hard (label) disagreement.

Performance Penalty

$$\phi_\epsilon(\mathbf{z}) = \exp(-\Delta(\mathbf{z})^2 / 2\epsilon^2)$$

A Gaussian boundary softly keeps candidates within the Rashomon threshold.

Combined Fitness

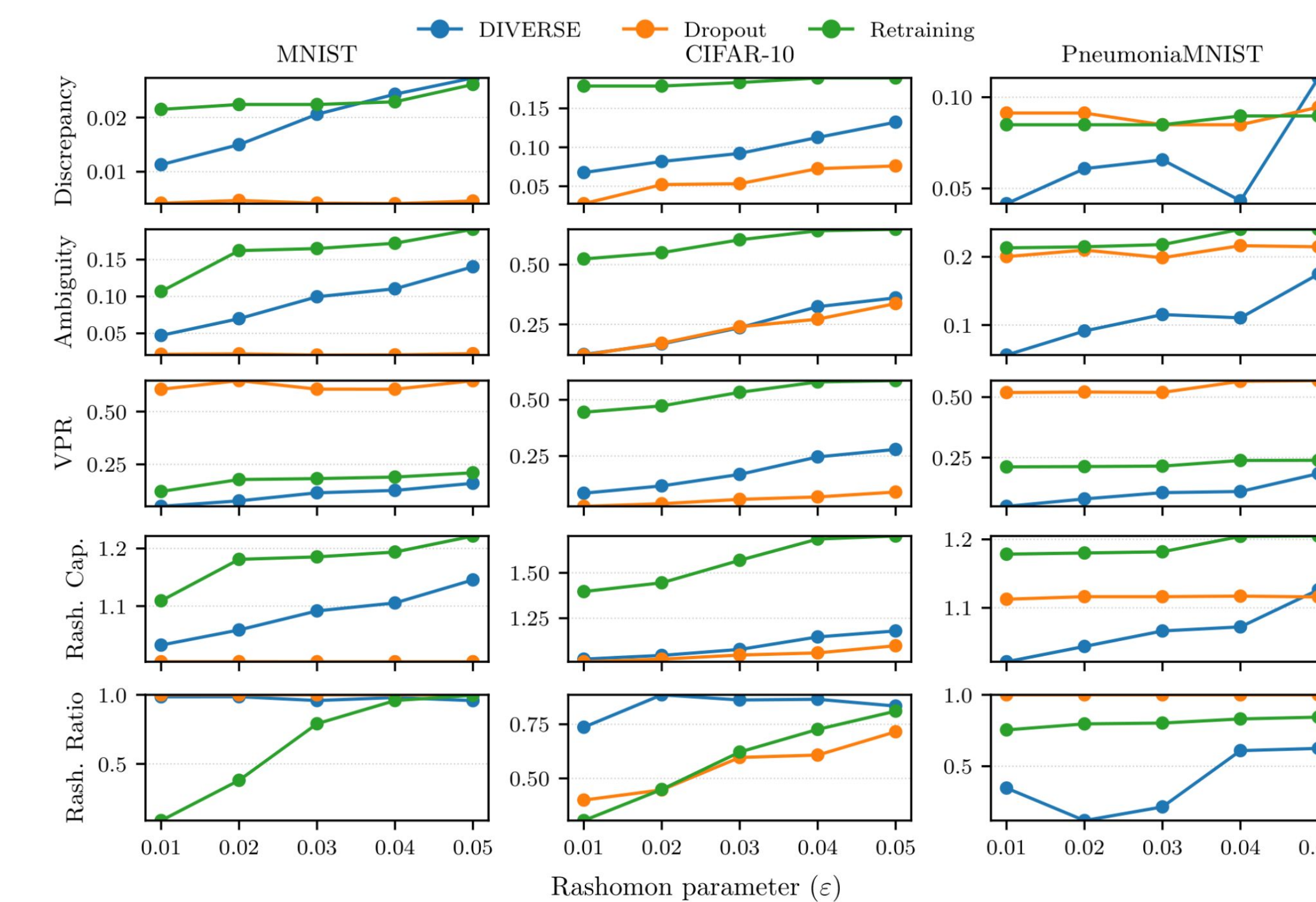
$$\mathcal{F}(\mathbf{z}) = \text{Div}_\lambda(\mathbf{z}) \cdot \phi_\epsilon(\mathbf{z})$$

4. Validate Rashomon Membership

CMA-ES searches over the training set, so Rashomon membership is evaluated separately on a held-out validation set using the same ϵ -threshold. This ensures that candidates genuinely satisfy the Rashomon constraint, rather than simply fitting the data the search was optimised on.

3 Results

How DIVERSE Are the Results?



Diversity metrics and Rashomon Ratio across Rashomon tolerances ϵ . DIVERSE consistently outperforms dropout on MNIST (MLP) and CIFAR-10 (VGG-16), approaching retraining performance, with diminishing gains on PneumoniaMNIST (ResNet-50).

Runtime Efficiency

	MNIST	CIFAR-10	PneumoniaMNIST
Retrain	00:29:39	03:17:26	02:09:00
Dropout	00:00:30	00:01:58	00:01:32
DIVERSE	00:00:50	00:02:11	00:01:49

Runtime (hh:mm:ss) to obtain 162 candidate models under the same Rashomon constraint.

Key Findings

- Efficient Rashomon Set Exploration**
Practical for large-scale multiplicity research in Neural Networks
- Extends to Vision Transformers**
Preliminary ViT experiment on CIFAR-10 motivating further study
- Layerwise Analysis**
Shows diversity is localised, enabling targeted future search

